Truong Gia Hy

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EDUCATION

Royal Melbourne Institute of Technology — Ho Chi Minh City, Vietnam
Bachelor of Information Technology — *2023 — present*
Fachhochschule Aachen — Aachen City, Germany
Bachelor of Applied Chemistry — *2021 — 2023*

PROFESSIONAL EXPERIENCE

Software Engineer Trainee, *(EON TECH)* — *12/2025 - 06/26*

- Developed a mobile application services using React Native and TypeScript, developed mobile UI/UX, collaborative workflows with Git.
- Contributed to building an Admin Content Management System (CMS) for managing mobile app data, integrating backend RESTful APIs.
- Implemented a Legal RAG AI-chatbot system, integrating legal sources from ThuVienPhapLuat reduce the responses' hallucinations.

Collaborative AI/ML Researcher, *(The Neurone Group)* — *09/2023 - present*

- Engineered CoT prompts, built LLM-based Chatbots, established PostgreSQL psycopg2 database with Docker for Text-to-SQL Agent.
- Experienced in supervised learning/unsupervised learning, applied Qwen2.5vl in reasoning images, research about CNN, Transformer.
- Researched about cryptographic techniques (e.g., AES, RSA, ECC) in enhancing data security and privacy within healthcare systems.

RESEARCH PROJECTS

Sai Gon Tam Duc mobile app: A booking clinic dental app — *EON TECH*

- A booking mobile app allows customers to book appointments for several dental services in several branches, which locates in Vietnam.
- Intergrated Frontend using Typescript, React Native and Expo Go via Figma design, used components for reusing in several code files.
- Enhanced UI/UX in both Android version and IOS version, created color theme dark and light mode for being suitable in devices system.

Sai Gon Tam Duc web admin: CMS Admin application — *EON TECH*

- A website for admin account only to manage the mobile app system and the booking information across the entire dental clinic system.
- Performed CRUD operations for multiple modules, products, announcements, branches, and personnel across the dental clinic system.
- Conducted testing backend APIs via Postman, conceptualized and implemented intuitive admin interfaces to optimize the overall usability.

PinSpots — *Github Link*

- A native Android application using Kotlin, Jetpack Compose, and MVVM, enabling users to discover and manage locations in the city.
- Implemented Firebase and Room Database for authentication, offline local storage first, allow multi-user collection sharing each other.
- Utilized Google Maps SDK for spot location search, interactive map visualization, and add accurately address-to-coordinate geocoding.
- Connected to the Open-Meteo REST API via Retrofit to provide real-time local weather conditions and forecasts for saved locations.

BlooMart: An E-commerce Website — *Github Link*

- An e-commerce platform allows customers to view and buy various products via Momo app from vendors and be delivered by shippers.
- Created the database with MongoDB Atlas, used Mongoose for schema modeling, and integrated it into a Node.JS and Express backend.
- Coding Backend using Node.JS, connected to Frontend via React Framework and axios, created interactive web interface using SCSS.
- Applied Agile methodologies with Jira to track weekly progress and manage timelines, ensuring each sprint completion rate over 84%.

Wander: A Tourism Forum Website — *Github Link*

- A platform allows users to share discussions about various tourist destinations and services, provides booking and payment experience.
- Designed the database using MongoDB Atlas, incorporating Mongoose for schema modeling and integrating it into a Node.js backend.
- Coding Backend using NodeJS, connected to Frontend via EJS Framework, created an interactive web interface using JS and CSS.

SLM Agent: Small Text-to-SQL Agents for Efficient Financial Data Retrieval and Strategization — *Github Link*

- Investigated 10 small language models ($< 12B$) to automatically generate SQL queries and natural language for information retrieval.
- Processed 4 financial datasets with over 10,000 observations and 12+ columns, pulled SLMs from HuggingFace, Ollama, ChatGroq.
- Research on NLP metric like RAGAS (Factual Correctness: F1, Precision), Cosine Similarity, BLEU, ROUGE to evaluate models output.
- Build ground-truth SQL queries pipeline using Groq API with Llama 3.3-70B model, manually evaluate the generated SQL queries.

PUBLICATIONS

[1] Ngan Nguyen Anh Nguyen* and Hy Truong. "Small Language Models for Efficient Domain Data Retrieval Using Structured Query Language from Natural Language". In: *SSRN* (2025). URL: https://doi.org/10.2139/ssrn.5909763.

TECHNICAL SKILLS

Code Language: *HTML, CSS, JavaScript, Java, ReactJS, ReactNative, Kotlin (Jetpack Compose), TypeScript, NodeJS, Git*

Digital Tools: *Canva Pro, PowerPoint, LaTeX, Figma, ChatGPT 4o, ClaudeAI, Gemini, Tableau, Jira, Notion*

Languages: *English (IELTS 6.5), German (A2), Vietnamese (Native)*